WORKED SOLUTION

STEP I 1990, Question 2

Pure • Roots of Unity Filter

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q2

This question has a full guided treatment in *Beyond STEP*, Section 5.2, *The Roots-of-Unity Filter*: https://beyondstep.keithmcnulty.org/ch05/roots-filter.html

The key idea

The cube roots of unity act as a filter. When we expand $(1+\omega^r)^n$ by the binomial theorem and add the results for $r = 1, 2, 3$, each binomial coefficient $\binom{n}{j}$ gets multiplied by $1+\omega^j+\omega^{2j}$, which is 3 when 3 divides j and 0 otherwise. So the sum keeps exactly the coefficients we want. The other way to evaluate each $(1+\omega^r)^n$ is to write $1+\omega^r$ in modulus–argument form and use de Moivre's theorem.

Solution

The cube roots of unity

Since $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, we have $\omega^3 = \mathrm{e}^{2\pi\mathrm{i}} = 1$ but $\omega \neq 1$. Then

$$0 = \omega^3 - 1 = (\omega - 1)(1 + \omega + \omega^2),$$

and since $\omega - 1 \neq 0$ we may divide by it to get $1 + \omega + \omega^2 = 0$.

It follows that $1 + \omega^2 = -\omega$. Multiplying by $-1 = \mathrm{e}^{-\mathrm{i}\pi}$ rotates through $-\pi$, so

$$1 + \omega^2 = \mathrm{e}^{-\mathrm{i}\pi}\,\mathrm{e}^{2\pi\mathrm{i}/3} = \mathrm{e}^{-\mathrm{i}\pi/3}.$$

So $1 + \omega^2$ has modulus 1 and argument $-\frac{\pi}{3}$. (As a check, $1 + \omega^2 = 1 + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$.) In the same way $1 + \omega = -\omega^2 = \mathrm{e}^{\mathrm{i}\pi/3}$, the complex conjugate.

The identity

First way: the binomial theorem. For $r = 1, 2, 3$,

$$(1+\omega^r)^n = \sum_{j=0}^{n} \binom{n}{j} \omega^{rj}.$$

Adding the three cases,

$$(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n = \sum_{j=0}^{n} \binom{n}{j} \left(\omega^j + \omega^{2j} + \omega^{3j}\right).$$

Ask yourself What is $\omega^j + \omega^{2j} + \omega^{3j}$ for a general integer j?

Since $\omega^{3j} = 1$, the bracket is $1 + \omega^j + \omega^{2j}$, and it depends only on j modulo 3, because $\omega^3 = 1$.

- If j is a multiple of 3, then $\omega^j = 1$ and the bracket is $1 + 1 + 1 = 3$.
- If $j \equiv 1 \pmod 3$, then $\omega^j = \omega$ and $\omega^{2j} = \omega^2$, so the bracket is $1 + \omega + \omega^2 = 0$.
- If $j \equiv 2 \pmod 3$, then $\omega^j = \omega^2$ and $\omega^{2j} = \omega^4 = \omega$, so again the bracket is 0.

So only the terms with $3 \mid j$ survive, and

$$(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n = 3\left[\binom{n}{0} + \binom{n}{3} + \cdots + \binom{n}{k}\right],$$

where k is the largest multiple of 3 that is at most n.

Second way: modulus and argument. From the first part, $1 + \omega = \mathrm{e}^{\mathrm{i}\pi/3}$ and $1 + \omega^2 = \mathrm{e}^{-\mathrm{i}\pi/3}$, and $1 + \omega^3 = 2$. By de Moivre's theorem,

$$\begin{aligned}(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n &= \mathrm{e}^{\mathrm{i}n\pi/3} + \mathrm{e}^{-\mathrm{i}n\pi/3} + 2^n \\ &= 2^n + 2\cos\left(\frac{n\pi}{3}\right).\end{aligned}$$

Comparing the two evaluations and dividing by 3,

$$\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots + \binom{n}{k} = \frac{1}{3}\left(2^n + 2\cos\left(\frac{n\pi}{3}\right)\right),$$

which is the required result.

The case $n = 25$

The largest multiple of 3 not exceeding 25 is 24, so the first sum is exactly the identity with $n = 25$. Now $\frac{25\pi}{3} = 8\pi + \frac{\pi}{3}$, so $\cos\frac{25\pi}{3} = \cos\frac{\pi}{3} = \frac{1}{2}$, and

$$\binom{25}{0} + \binom{25}{3} + \cdots + \binom{25}{24} = \frac{1}{3}\left(2^{25} + 1\right) = \frac{33554433}{3} = 11184811.$$

The second sum

Here $n = 24$ and we want the coefficients $\binom{24}{j}$ with $j \equiv 2 \pmod 3$. Group all the coefficients $\binom{24}{j}$ by the remainder of j on division by 3:

$$S_0 + S_1 + S_2 = \sum_{j=0}^{24} \binom{24}{j} = 2^{24},$$

where S_m is the sum of $\binom{24}{j}$ over $j \equiv m \pmod 3$.

The symmetry $\binom{24}{j} = \binom{24}{24-j}$ pairs these up: if $j \equiv 2$ then $24 - j \equiv -2 \equiv 1 \pmod 3$, since 24 is a multiple of 3. So $j \mapsto 24 - j$ matches the terms of S_2 one-to-one with the terms of S_1, and $S_1 = S_2$.

The identity with $n = 24$ gives S_0: as $\cos 8\pi = 1$,

$$S_0 = \frac{1}{3}\left(2^{24} + 2\right).$$

Hence

$$S_2 = \frac{1}{2}\left(2^{24} - S_0\right) = \frac{1}{2} \cdot \frac{2 \cdot 2^{24} - 2}{3} = \frac{2^{24} - 1}{3}.$$

Halving the given value, $2^{24} = 16777216$, so

$$\binom{24}{2} + \binom{24}{5} + \cdots + \binom{24}{23} = \frac{16777215}{3} = 5592405.$$

Checks

- Small cases: for $n = 3$ the left side is $1 + 1 = 2$ and the right side is $\frac{1}{3}(8 + 2\cos\pi) = 2$; for $n = 4$ it is $1 + 4 = 5$ against $\frac{1}{3}\left(16 + 2\cos\frac{4\pi}{3}\right) = \frac{1}{3}(16 - 1) = 5$.
- Divisibility: $2^{25} + 1$ and $2^{24} - 1$ must both be multiples of 3, and they are, since $2 \equiv -1 \pmod 3$.
- We summed the binomial coefficients directly by computer: the identity holds for every n up to 199, the first sum is 11184811 and the second is 5592405.

Common slips

- Giving the argument of $1 + \omega^2$ as $\frac{\pi}{3}$: it lies in the fourth quadrant, so the principal argument is $-\frac{\pi}{3}$ (equivalently $\frac{5\pi}{3}$).
- Forgetting the case $r = 3$, which contributes the 2^n.
- In the second sum, trying to use the formula directly: it only gives multiples of 3, so another idea (here, symmetry) is needed for the remainder-2 terms.

Beyond STEP

This trick is called a *roots of unity filter*, and it is a first example of the discrete Fourier transform: averaging $f(\omega^r)$ over the mth roots of unity picks out the coefficients of a polynomial f in a chosen residue class modulo m. The same averaging over a group of symmetries reappears in the character theory of finite groups and in counting arguments such as Burnside's lemma. For more on why complex numbers make such calculations natural, see Needham, *Visual Complex Analysis*.

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